

TP1 filtering

First exercise

1) Go to your working directory (“open folder” in VSC)
Open a new jupyter file (“new file”)

in the first code block, import all the libraries

```
import numpy as np #library to manipulate arrays
import matplotlib.pyplot as plt #library to plot figures
import scipy.signal as sci #signal processing library
%matplotlib qt
```

run that block

2) in a second code block

load the mystery signal

```
s=np.loadtxt("mysterySignal.txt")
```

plot the temporal vue

compute and print the mean

compute and print the power

what else can be done ?

3) quantize this signal over 8 bits (sq)

plot s and sq

plot the error

what is the expected SNR?

Measure the SNR

check the probility distribution